

Instructions in preparing an application

Applicants are required to use this template to submit their Flagship program. Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. All indicated chapters are mandatory unless otherwise indicated. You must respect the indications given in each chapter in order to adhere to the expected content. A word count is clearly indicated for each section. **If you do not adhere to the maximum amount of words, your application will not be processed.** The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent.

We recommend that you do not send in your submission at the last minute in order to avoid technical difficulties. No submission will be accepted after the deadline, **March 15th at 13:00 CET**. Applications should be sent in PDF format to **Flagships@erasmusmc.nl**.

Title: Tailoring Technologies for a Healthy Neighbourhood (The Healthy Neighbourhood Project)

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship [Main Lead/Lead/ Applicant]</i>
Samantha Copeland (SC)	TUD	TPM Values, Technology and Innovation	s.m.copeland@tudelft.nl	Tenure track		Main Lead
Lieke Oldenhof (LO)	EUR	School of Health Policy & Management	oldenhof@eshpm.eur.nl	Tenured		Lead
Lex Burdorf (LB)	EMC	Public Health	a.burdorf@erasmusmc.nl	Tenured	Researcher	Lead
Ingrid Mulder (IM)	TUD	IDE Human-	i.j.mulder@tudelft.nl	Tenured		Applicant

		Centered Design				
Clarine van Oel (CvO)	TUD	Architecture and the Built Environment	c.j.vanoel@tudelft.nl	Tenured		Applicant
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Debora Antunes (DA)	EUR	ESHCC Media and Communication	debora.antunes@eshcc.eur.nl	Other: Lecturer		Applicant
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Famke Mölenberg (FM)	EMC	Public Health	f.molenberg@erasmusmc.nl	Other: Postdoctoral Researcher	Researcher	Applicant
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Wilma Jansen (WJ)	EMC & City of Rotterdam	Maatschappelijke Ontwikkeling / Social Development	w.jansen@Rotterdam.nl	Other: postdoctoral researcher	Researcher	Applicant

Corresponding Lead: Samantha Copeland (TUD s.m.copeland@tudelft.nl)
 Planned start date of seed funding: 1-09-2022
 Primary Convergence theme: *Technology Supported Transitions in Health(care)*
 Secondary Convergence theme (if applicable): *Improving Health Journeys*
 Tertiary Convergence theme (if applicable): *Human Centered Technology and AI for Health*
 Quaternary Convergence theme (if applicable): Choose an item.
 Keywords: *healthy neighbourhoods, living environment, equity, stakeholder networks, tailoring technologies, self-sufficiency, positive health*

1.b Summary (max 300 words)

Neighbourhoods are frequently targeted as sites for implementing technology supported health interventions to improve health and reduce health inequalities. With some notable exceptions, these tend to treat and fix the neighbourhood as a stable determinant when measuring health outcomes. Consequently, we diminish intervention potential since we lack insights into how ‘the healthy neighbourhood’ is dynamically constructed and negotiated over time in daily interactions between inhabitants, local stakeholders, governments and service providers, as well as technologies and physical environments. Furthermore, we lack insights into the access, uptake and effectiveness of these novel technologies to improve lifestyles of vulnerable populations. These populations typically stay behind the technology- and government-driven digitization of healthcare, health promotion and support, resulting in increasing health disparity. This project begins by assuming that creating a ‘healthy neighbourhood’ means creating a bottom-up driven stable network of socio-material relations in which technologies and individuals can be jointly mobilized and supported by local stakeholders. We see the neighbourhood as a living lab for generating understanding about and improving technology based health interventions. In this innovative proposal, we address the health potential of interactive technologies within urban neighbourhoods. The abundance of suboptimal health technology creates a problem rather than a solution for a neighbourhood in the current situation. By integrating and tailoring health technology to the specifics of the neighbourhood, the negative effects can be reversed. We bring together diverse experts in human-technology interaction from three institutions with the aim to increase access to and uptake of tailored technological health interventions at the local level. This will improve long-term integration of preventative health measures and improved attitudes toward and access to health related technologies that individuals and groups can use in their everyday lives and that are supported by local stakeholders.

1.c Public Summary (max 300 words)

How can we design technologies that cultivate both health and healthy neighbourhoods?

Technological interventions to improve health such as apps, digital communities and research tools often focus on either individual health, population health or healthy, supportive local environments, and not the connections between them, whereas all are needed for a ‘healthy neighbourhood’ goal to be achieved. We define healthy neighbourhoods as places where the people who live there have strong relationships, both between themselves and with their environment (social, organizational, cultural and built environment that makes up the ‘place’

where they live). Health technologies designed to promote health within the living environment require uptake from users, which can be ensured via co-creation and the building of the kinds of relationships that will sustain tech-based practices. Therefore, cultivating both health and healthy neighbourhoods is a way to ensure technologies are likely to be effective and relevant to the people and places where they are implemented.

Our multidisciplinary team incorporates social sciences, media studies, humanities, medicine, and community organizations, taking experimental and co-creative approaches to the design and implementation of health technologies. We develop theory and a guide for practice that will enable the tailoring of technologies for the needs of particular neighbourhoods and provide decision-making support for stakeholders, government and service providers. By developing a concept of the Healthy Neighbourhood throughout the project, we draw on increasing knowledge about the social determinants of health and give substance to how we might connect the various domains of positive health, while implementing technologies toward transitions from the hospital to community health, and from reactive to proactive approaches to health promotion.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

The consortium brought together with this flagship program is truly interdisciplinary and will address several aims of Convergence. It will provide theoretical and empirical bases for better uptake of existing healthcare technologies and implementing novel ones within neighbourhoods in need of health interventions. This will result in improvements in design and application of technologies that monitor and assist in health-conducive lifestyle and uses of the built environment, beginning with youth and extending approaches to include the full lifecycle. We contribute directly to the main goals of Convergence: life-long health and well-being by supporting health-conducive relationships between community members and institutions within the neighbourhood; Attending to socio-economic equality in health and wellbeing by reducing disparities at the level of the neighbourhood by tailoring technologies to local needs of the community⁵⁹; Implementing and refining technologies that diagnose and recommend preventative measures for health and wellbeing throughout the lifecycle. These goals will be accomplished by 1) gaining insights into the actual uptake of health technologies within neighbourhoods⁶⁰ and 2) tailoring technology to individuals and local stakeholders, to develop principles for identifying and addressing the unique needs of different neighbourhoods, to implement technologies in ways that generate healthy neighbourhoods as well as healthy and self-sufficient individuals.

A high level of integration is required for developing an effective transdisciplinary approach to both the Healthy Neighbourhood (HN) framework we develop. We implement it in practice through co-creation, thereby ensuring the principles developed and results obtained accommodate the goals of improving equity in health and promoting the transition toward proactive, localized care, promotion and support, by improving user engagement and strengthening supportive stakeholder networks via decision-making support.⁶¹ *This integration is our focus: optimising these processes and generating the best outcomes at all scales is as*

important to making transitions toward proactive approaches and increased access to care within the living environment as developing new technologies can be.

2.b Synergy between disciplines

Researchers involved have diverse expertise in methods from anthropology, bioethics, public health, design, ethnography, social transformation theory, storytelling and nudging, and lifestyle coaching, and who currently work with eHealth, responses to the built environment, approaches to sport and health, public health and health promotion, equality of access and health outcomes, uses of social media and discussion forums, and participatory ethics. This diversity of methods and expertise brought together enables a comprehensive framework for the HN and a robust picture of the potential of technology to generate health-conducive relationships. Past decades are marked by a technological focus resulting in innovative eHealth applications such as monitoring systems, self-management tools, digital revalidation and prevention programs, involving domains mainly consisting of medical- and technological sciences. However, the reach and uptake of these technologies by targeted end-users, e.g. patients, people at risk, is low on long-term usage.^{1, 2, 3} Adding behavioral scientists to the medical and technological scientific domains improves uptake and reach of health innovations but is still not optimal, demonstrated by growing health disparity and short term usage of eHealth. Humanities is included here to collaborate with medical, technological and behavioral scientists to provide context analysis and intervention opportunities. In our project the research perspective is shifted from top-down medical norms and technological interventions to bottom-up approaches of human and social needs to which existing and future technology can be tailored. The consortium has come together under this common interest in the importance of (a) building well on approaches already being taken and (b) developing a theoretical foundation for methods in design and implementation, grounded in co-creation and attention to local needs and practices.

We will optimize and further technologically-mediated relationships between people-people and people-environment, by: (i) studying and mapping the potential of existing technologies (eg. social media for people-people relations; XR in people-environment relations), (ii) facilitating and evaluating the adoption of technological interventions by engagement-oriented methods like storytelling, building relationships among community members and between the community and healthcare resources over the long-term, (iii) Cocreative tailoring and evaluating of local technology governance and use, to build strong and supportive neighbourhood networks of stakeholders.⁶² In addition, we use Theories of Change⁶³ towards a transferable methodology for further enhancement and upscaling of technological health interventions⁶⁴, while taking into account place-based characteristics. This generates basic principles that can be used in the later years of our project and beyond, toward the (iv) cocreative tailoring of technologies toward social as well as physical and mental health.

Overarching workgroups integrate a number of disciplinary approaches, perspectives, expertise and focus. In particular, workgroup A takes a transdisciplinary approach by employing co-creation in the early stages of the research as well as ethnographic study of the context in which technologies are currently being employed. Close ties with Rotterdam municipality and local institutions enable Group B to work with stakeholders—community members, families and youth, as well as local institutions such as schools, sporting clubs and service providers in the social and health domain (see §4)—to craft a clear idea of the most important elements of the HN concept and how it might be employed through design and collaboration, and/or exploratory living labs. Group C in turn takes an interdisciplinary approach, bringing together researchers in the

consortium to reflect on the scientific implications of the living labs, principle-evaluation and theory-formation, at regular intervals throughout the project (see §3).

2.c Long term sustainability plan

One risk to the success of our research aims is the feasibility of developing generalizable principles for attending to the specific needs of residents of neighbourhoods through healthcare technologies. It is possible that some health goals do not align well with the goal of a healthy neighbourhood; it is also a matter of debate whether the neighbourhood is the best unit of analysis for understanding the needs of urban individuals, or what roles social cohesion, social resilience and local services might play in engaging local groups, for instance.⁴ For these reasons we take a participatory, co-creative approach for developing our healthy neighbourhood approach, and a directed, integrative approach to developing and implementing our design principles and integrating our results. Tackling these challenges through diverse disciplinary lenses together will establish joint research lines with long-term viability, via joint publications and workshops.

Further, the consortium activities will solidify partnerships with Rotterdam municipality, across the three Convergence institutions, EMC, EUR and TUD, with other Convergence initiatives, namely Resilient Delta and other flagships with which consortium members collaborate, and also with Living Labs by the Hogeschool -Den Haag, -Rotterdam, -Leiden and InHolland. The flagship will co-ordinate where possible with other long-term initiatives, such as Healthy Start and the Digital Buddy projects. Our intended outcomes are progressive, and consequently open-ended; while we hope to generate and pursue several methods for tailoring technologies within the proposed timeline, we fully expect that our work together will open up additional avenues for improving partnerships between neighbourhood inhabitants, service providers and other stakeholders, researchers, technology developers and designers, and local organizations and institutions, and thereby the extension of the scope, implications and potential of this research.

Specific efforts already in place to expand and extend the flagship program; we have sufficient awarded matched funding from consortium-led projects for the first two years (first table) and projected continuing funding (second table):

Funding agency/grant	Funding to match
ZonMW (VV – Healthy Storytelling -2022)	€ 50 000
NWO (VV – Enduring Rewards -2024)	€ 1 100 000
Benefit4All (Medical Delta, VV – eHealth design -2023)	€ 400 000
T-AP SI (CvO – uVITAL -2023)	€ 60 000
NWO (CvO – VR-Renovate -2025)	€ 60 000
Funding agency/grant	Matched funding
EU Horizon (VV Obesity Prevention 2023-27)	€ 10 000 000

Horizon NEB (IM 2022-25)	€ 50 000	
NWO-KIC (CvO Learning Community 2022-28)	€ 900 000	
NWO-KIC (VV, SC, IM, LO, WJ, Medical Delta 2022-2028)	€ 2 000 000	

2.d Visibility, attracting talents and new partners

To succeed in our impact plan, we attend to the visibility of our HN approach and encourage contributions from global experts in our various fields, by holding international as well as local workshops and joint publications. Within the first three years, we will publicize our proposed model and build a virtual network of collaborators around discussions and resources relevant to making it more robust and transdisciplinary and to fully explore the scope of potential application. The postdoctoral researcher we employ will be devoted in part to the collaborative effort itself, ensuring that publications take advantage of the overlap between otherwise diverse methods and disciplinary focus, and creating opportunities for out-of-the box thinking and space for exploratory work by organizing retreats and writing workshop days on a regular basis, intended to maximise potential connections within the consortium.

Presentations and panels that center on this topic will be appropriate for international conferences such as EASST, 4S, EPH, IAPS.

Specific and ongoing plans to attract funding and connect with new partners in the near future, as well as recently funded projects lead by consortium members (see tables in §2c) include: User-valued Innovations for Social Housing Upgrading through Trans-Atlantic Living Labs (uVITAL), Social Innovation program; KIC & Learning Communities; Enduring Rewards: Maintaining a lifestyle change by eHealth supported dynamic personalized incentives; Benefit4All - eHealth design for people with low social economical; Obesity prevention throughout life; lighthouse demonstrator grants for the New European Bauhaus initiative. These and other ongoing and intended applications for funding will support the broader consortia activities while Convergence funds are essential for strengthening the network among us and enabling truly transdisciplinary and robust results that keep to our goal of optimizing current work toward better health.

2.e Long term societal impact

Concrete, long-term impact of the project will be in the areas of theory, data, experimental results, policy recommendations, implemented policy changes, and technology development and design.

This flagship focuses on how people interact digitally and physically within their neighbourhood, and the implications this has for health and health technologies. Interpersonal neighbourhood relationships can be viewed through a combination of data-driven digital means and understandings of situated activities - the key challenge is knitting together overlapping layers of relations that provide context for healthy interaction. We thus investigate: how can digital phenotyping approaches be used to characterise interactional qualities of neighbourhoods?; what do residents' digital footprints show of their connections?; how can we discover and support interactions that lead to healthy outcomes?; how can existing eHealth applications be adapted and positioned in networks to reach the right (more) people?; how can local stakeholders support

and strengthen inhabitants, generate more impact and improve health at the neighbourhood level?

Known pathways for impact include a continuing and further developing collaboration with Rotterdam municipality to extend the existing data dashboard utility for youth towards the lifecycle, as well as tailoring interventions and improving decision-making processes better toward local data needs and knowledge on evidence-based interventions.⁵ We can further build on the results of an ongoing mixed methods evaluation study (Promising Neighbourhoods) directed at health, safety and talent development among youth (Boelens 2019). This also corresponds with the aims of collaborating projects (eg. Digital Buddy) to expand the scope of application of current apps beyond youth and young families. Such optimization thus is already desired and our project provides an ongoing platform for assessing effectiveness and identifying knowledge gaps, and contributes to efforts to extend the usability of dashboards and apps to include all age groups, to improve health, safety, wellbeing, self-sufficiency, equity and access within the living environment.

Initiated projects by consortium members will form novel partnerships in research and through joint efforts toward the dissemination of results. For instance, we are simultaneously applying for the NWO-KIC Access to care in the living environment consortium grant, which includes establishing a partnership with additional institutions through the Medical Delta collaboration, including LUMC and Leiden municipality, as well as the Hague. Collaborations within the consortium will have impact on the expansion and design of technologies such as the Digital Buddy, but also the gaming app of the Torrie project, as well as offering design guidance for the implementation of social media and community apps (burenapps), the use of cross-reality (XR) to facilitate communication and enhance empowerment, and municipality tools within local contexts toward health goals.

Further, the combination of expertise in the consortium will ensure optimal uptake and engagement. For instance, the current proposal builds upon ongoing research⁶ that cities – and other urban stakeholders require the capability to design for unpredictable and rapidly changing futures, in a citizen-centred way, but that existing methodologies in design and governance often fall short to foster such long-term, complex, and non-linear processes of systemic change. The work takes stock of the recently finished DESIGNSCAPES project that supported 100 mission-driven local innovation pilots to investigate the transformative power of design in societal challenges. Two leading Dutch pilots that focused on Designing for healthier citizens have been instrumental for the setup of this proposal.⁷ The cross-pilot understanding of the tools, processes, instruments, and techniques in design-enabled innovation as well as the transferable methodology for building capacity to achieve a better uptake, further enhancement and scaling of the local experiments (living labs for tailoring technologies) will be leveraged.

Finally, the education of at least 2 PhDs by committees formed from the three institutions and of differing expertise will contribute to developing a new generation of Convergence-oriented researchers. Impact will also be realized at the BSc and MSc levels within the three institutions, through novel and interdisciplinary projects and assignments: SC will make contributions to a new MSc elective course in health systems at TPM-TUD, ensure the addition of in-class discussion and projects about ethics issues raised when introducing health technologies into communities for MSc students in the elective (available to external students and already attended by Clinical Technology students, eg.) WM1402TU at TUD; and the flagship will provide topics for assignments and discussion in TB212A, Philosophy, Ethics and Safety Science (BSc TPM-TUD). JK will include at least three MA thesis students and offer internships for BA thesis students (IBCoM) for the evaluation process of city labs (possible programmes are Media & Society, Media & Creative Industries at ESHCC). LO will incorporate topics for assignments and action research in the

interdisciplinary courses Governing Healthy Cities (EUC and ESHPM) and Care and Wellbeing (ESHPM).

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

This proposal is innovative in that we focus on how to tap the full potential of existing and future healthcare technologies to aid in the transitions from reactive to proactive care, and from the hospital to the community. Our project hones in on current efforts and provides a framework for moving forward in an effective, inclusive and engaging way. It is also therefore challenging: in true Convergence fashion, we bring together multiple disciplinary perspectives and experience to ensure the framework and guidance on how to improve the use of healthcare technologies in urban neighbourhoods are robust and ready to use. There are increasing calls to transfer the burden from formal institutions to community organizations and to encourage self-sufficiency in patients. This and other aims of recent legislation require greater integration of stakeholders, authorities in medicine and politics, and end-users, patients and participants in proactive health measures. Our project as well as the municipalities and groups we work with will therefore benefit from our collaborations with local stakeholders and service providers, as well as patients, to contribute directly to their efforts with our results.

For the transition of health promotion and support into the living environment of neighbourhoods to be effective, it will require long-term effective take up of technology implementation strategies by the target groups, and also avoid those living in disadvantaged urban neighbourhoods from being overwhelmed by de-contextualized technological novelties resulting in poor uptake of them and stigmatization. We argue there is a need for a place-based approach derived from a sense of the neighbourhood that allows for flexibility, specificity, and diversity within the group and locale upon which an intervention focusses. Further, the need for expansion of applications to the full lifecycle and for capturing groups otherwise missed requires the optimization of current uses. To make health technology interventions more effective, a transdisciplinary approach is required that facilitates a bottom up integration of perspectives from local stakeholders such as the municipality and service providers, health technology developers, and inhabitants. Bringing together such diverse perspectives is a difficult task that our model can ground more effectively by starting with the basic components of a healthy neighbourhood and generating rather than imposing our HN model through practice.

Finally, we hope as a group to answer the questions: How does technology create new opportunities for relationship-building? Can we consider the digital space an additional aspect of the healthy neighbourhood environment, and how? To make our long-term aims feasible, our living labs will also consider these broader questions, including intentional efforts to attend to the needs and enable engagement among potential users who tend to be neglected when introducing new interventions (eg. low-SES groups, but also those less familiar with the digital environment - what benefits can we say these technologies bring to them?). And, how might technology assist local government and local stakeholder networks in better tailoring and targeting their health

policy, health promotion and support activities to the needs of inhabitants and monitor the impact of measures and interventions?

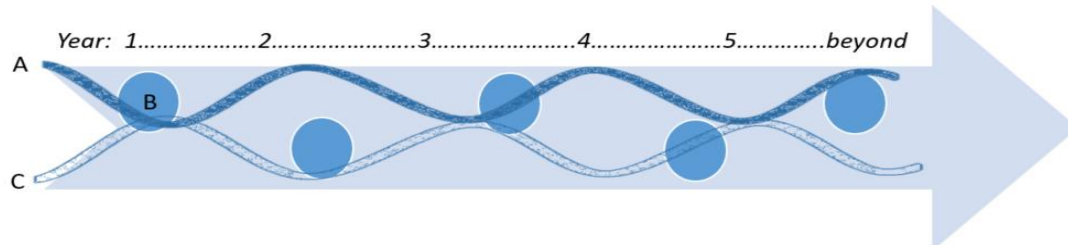
3.b Aims and feasibility

We have the following objectives and plans to enable their achievement:

- Our overarching objective is to produce a transdisciplinary approach under the common concept of a healthy neighbourhood (HN), to integrate health technologies in the local, organizational, social and physical, living environment, and principles for tailoring technologies toward these aims.
- The theoretical grounding for this approach will be developed via methods in philosophical inquiry and ethics (SC) as well as in local transformation processes (IM), ethnographic approaches to understanding local needs (LO), including digital ethnography (DA, JvS) and co-creative approaches to enable participant engagement and empowerment from the early stages of theory-formation and validation (FM, JK, CvO, RST, NV, VV). Developing new theory provides an excellent opportunity for PhD-level exploration, who will contribute literature review and exploration of ideas relevant to generating the HN framework itself.
- The HN framework provides the means for determining which interventions may be best for which neighbourhoods (groups/locales) and for better (longer, more active) uptake of technology via user engagement and co-creation of the approach. To ensure uptake, we partner with Rotterdam municipality in data dashboard optimization for better tailoring to local data needs (WJ), and utilize connections with Medical Delta's 'eHealth and Selfmanagement' research group (VV), as well as other ongoing projects (RST, VV) and by tapping into team expertise in public health policy pathways (FM, LB, LO).
- Experimental living labs are a key method for evaluating principles for the HN framework and refining processes for user engagement, empowerment and uptake of evaluations by service providers. Expertise in designing, implementing and assessing the results of such living labs is demonstrable in the team (eg. CvO), including experience initiating CARE Lab Rotterdam (LO), the European Network of Living Labs (IM), the mixed methods evaluation of the Promising Neighbourhood project (WJ) and other past successes with citizen science (JK, FM, VV). Our collaboration with Medical Delta Living Labs will provide opportunities for comparisons and validation in additional contexts.
- Applications of HN principles toward tailoring technologies at the stages of design, implementation and establishing standardized intervention options for local stakeholders will draw on expertise from within and beyond the consortium. Technologies immediately available within the consortium for use in obtaining this objective include Digital Buddy and the Torrie project gaming app, as well as the processes involved in data dashboard utilization by the city of Rotterdam within neighbourhood stakeholder networks. Social media and gaming apps are readily available for use in experimental living labs during the project, including community apps, whatsapp groups and instagram. Further opportunities to innovate or further develop apps and similar technologies with partners familiar to consortia members will be pursued as the HN principles are developed through co-creation and assessed in practice, including game developers (eg. Ranje, TU Delft GameLab). Consortia members have further experience with digital diaries (JvS), storytelling technologies (VV), digital twin technologies (RST, VV), user-generated platforms such as SEEDS (WJ, FM), and cross-reality (CvO) that will lead to innovations

within diverse categories of technological interventions in the final years of the flagship and beyond.

3.c Work plan



Year 1-3: Framing the healthy neighbourhood as a combination of relationships between people, as well as with community organizations and institutions, through bottom-up participatory approaches (co-creation with community leaders, families and youth, especially in respect to priority-setting, improving long-term user engagement and uptake of appraisals) and theory development. Consequently, **Group A and the postdoc** initiate the project with a series of **full day workshops** to define our approach and basic conceptual common ground (2022/23). Together with the PhD students' **literature review**, the postdoc will develop a **resource base** for consortium use. By developing a **program theory**, the researchers of Group A also work to make explicit the connection between the HN and the health outcomes desired through technology as well as potential impact on health inequalities, user engagement and strengthening stakeholder networks. This impact will be evaluated and refined together with **Group C**. Thus co-creation work, involving local stakeholders and participants for whom the interventions will investigate apply, plays a key role from the beginning; these include youth, families, low-SES residents and service providers, as we optimise connections present in the projects of Group C researchers. **Ethnographic approaches** to the application of healthcare technologies in the living environment will further develop the HN framework around which other flagship activities are centered. Additionally, key challenges in terms of health and health inequities within and between neighbourhoods involved in living labs will be identified.

Year 2-3: The transdisciplinary work done in Year 1 will ground pilot applications (**Group B**) in Year 2 and for the remainder of the flagship, and these outcomes will be evaluated by **Group C** along with the postdoc researcher.

Group B, accordingly, includes **various technologies** that will be used to evaluate the theoretical and practical proposals regarding the HN approach developed by Group A. These will also be evaluated through an equity lens, providing insights into the access, uptake, and effectiveness of technologies for different population subgroups. Furthermore, we will provide insights into the mechanism of how these technologies impact the lifestyles of populations, and subsequently reduce inequalities between and within neighborhoods, as well as for positive changes in attitude and engagement toward and with technologies by endusers. **Living labs will be conducted** by Group C toward tailoring technologies and **determining basic principles** for this aligned with the goals forwarded by the HN approach, which Group A will in turn take up. This work contributes to and benefit from **ongoing projects**, eg. the extension of Healthy Storytelling (VV), Digital Buddy and KIC Digital Twin (VV, RST), uVITAL and VR-Renovate (CvO), and the ZonMw Sports During the Pandemic projects (JvS) (ie: Group B).

The postdoc completes their term by submitting a report including a review of potential options for scaling up and widening the scope of the project, including potential collaborations and applications for eg., in Rotterdam: the extension of dashboard data to include information and interventions across the lifecycle, to optimize the dashboard to better tailoring to local data needs, to optimize the assessment of effectiveness of intervention implementation, identify gaps (need-intervention-implementation).

Year 3-4: Having secured further funding for both ongoing projects and new projects generated by the collaboration, additional PhDs will be hired as well as a second postdoc; some overlap will enable a smooth transition, and the formation of a PhD group of both ongoing and new graduate students is supported.

Year 4-5: Establishment of the consortium and broader network around the HN framework and Tailoring Technology approach in the form of an ongoing hub for collaboration and research; an international, multidisciplinary conference under this topic will increase visibility of the work we have done as well as formalise the consortium at the center of a broadening network. PhD students will be cross-supervised by the three institutions. They will work eg., on the problem of tailoring technology, linking technological design with applications toward the healthy neighbourhood ideal with a focus on the built environment, as well as on narrative, storytelling and the lifecycle in the context of tailoring technologies toward health outcomes through social means. PhDs will all work together and organize consortium workshops. The postdocs will attend to the collaboration and impact assessment of the tailoring pilots by organizing the workshops in the first years to set up the HN framework and the terms for sharing results on an ongoing basis, and within the first two years perform and standardize the procedure for impact assessment and valorization of consortium results. Later postdoc hires will revise procedures where necessary, and consolidate the consortium's broader network to ground the project's extension beyond the 5 years (see §4d).

3.d Alignment of the program with relevant national and international research agendas

The flagship aims clearly align with the national agenda, particularly the aim to contribute to the health and self-reliance of citizens, and the transfer of care out of hospital centers and into communities in a proactive approach to health. Our initial research questions are aligned with the existing research agendas of the kenniswerkplaatsen and regional networks. Moreover, these questions will be further tailored to local needs together with local stakeholders.

For example, we will build on research that led to the digital Erasmus MC Life-Course Care platform as the 'Smarter Healthy App', launched in 2021. With this new technology lifestyle and medical care and practical information are integrated in digital patient journeys. (see www.Slimmergezond.nl). We follow the recent outcomes and potential for scaling up presented by the DESIGNSCAPES capacity building program. Further, The City of Rotterdam has identified 7 places ("Stadsprojecten") where urban regeneration will take place, modifying the neighborhood towards green zones to mitigate the effects of climate change while promoting health, and benefit from what has already been developed in previous Living Labs from the VR-Renovate and uVITAL projects. Finally, we intend to work alongside interventions already being implemented wherever possible and to seek further opportunities for doing so.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

BB-Beitske Boonstra: has expertise on active citizens and community initiatives in urban development and their interactions with (local) governments; worked at TNO Strategy & Policy on technological, process and social innovation; coordinator of the Academic Workplace Livable Neighbourhoods (Erasmus University & municipality of Rotterdam); conducting the project "Boundary spanning the sustainable city"; academic lead in the Rotterdam Resilient Delta initiative.

DA-Déborá Antunes: lecturer at the department of Media and Communication EUR; duties include supervision of BA thesis in Media & Health and MA thesis for tracks Media & Business and Media & Creative Industries; experience in conduction of digital ethnography and uses a biopsychosocial approach to health; focus on disability studies.

LB-Lex Burdorf: head of department of Public health Erasmus MC; observational and experimental studies in occupational health and public health, primarily on health promotion and participation, particularly how working and living conditions shape health trajectories through life; ample experience with field studies, and collaboration with various societal stakeholders.

SC-Samantha Copeland: Assistant Professor in Ethics and Philosophy of Technology at TU Delft; CauseHealth international network of clinicians and philosophers, developing educational resources for clinicians; expertise in bioethics, social resilience, participatory approaches to ethics in transformative and transdisciplinary contexts; co-PI on the Converging Ethics for Converging Technologies Impuls flagship, 2020-2022, Values Taking Shape Convergence Open Mind grant; supervises bachelor (BEP projects + manager TB212A) and MSc (5 projects + manager WM1402TU); PhD (1), and postdocs (3).³⁶

WJ-Wilma Jansen: postdoctoral researcher with city of Rotterdam, affiliated with Department of Public Health at Erasmus MC through hospitality agreement; research on prevention, health promotion and care use in youth; expertise in early detection, observational and evaluation studies on domains of lifestyle, overweight, mental health and wellbeing and care use among youth; is coordinator of the Regional Workplace Youth ST-RAW, supported by ZonMW to strengthen the knowledge-base and local learning environment to improve health, safety and development of youth; currently PI in generation R, involved in the European Horizon 2020 Citizen Science project SEEDS and PI in the ZonMW project Promising Neighbourhood that evaluates the "wijkprogramming" approach in Rotterdam.⁸⁻²¹

JK-Julia Kneer: Associate Professor for Media and Communication at Erasmus University, ESHCC; taught 80+ courses on BA and MA level, coordinator for BA and MA courses; supervised 80+ BA thesis, 30+ MA thesis, six PhDs; research focuses on the role of media and the impact of media on wellbeing and integration; has developed and tested several quantitative methodological approaches to investigate processes of empowerment of youth from highly diverse neighbourhoods; interdisciplinary projects including teachers, coordinators of integration programmes, coaches and schools: eg., "Homies: Neighborhood oriented media-based identities with empowering strategies" (Smart Urban Regions of the Future (SURF), together with Prof. Jeroen Jansz) and "Urban Media Engagement" (Erasmus Initiative Vital Cities and Citizen, together with Prof. Jeroen Jansz and Prof. Liesbet van Zoonen).²²⁻²⁵

FM-Famke Mölenberg: epidemiologist and postdoctoral researcher at the Department of Public Health at the Erasmus MC; affiliated with academic collaborative center CEPHIR. Her research is on preventing disease and reducing health inequalities; closely collaborates with policymakers and stakeholders; has a hospitality agreement with the City of Rotterdam; expertise in studying neighborhood interventions using quantitative methods. She is currently a member of the Erasmus Initiative Smarter Choices for Better Health (action line Health Equity) and involved in the European Horizon 2020 Citizen Science project SEEDS, to empower Rotterdam youth in deprived neighborhoods to create a change towards healthy lifestyle.²⁶⁻²⁸

IM-Ingrid Mulder: Associate Professor at TU Delft; director of Delft Design Lab Participatory City Making. Her background in policy, organization, and behavior sciences and cross-disciplinary experience at the national Top Technology Institute on Telematics anticipating future technologies impacting society has shaped her transdisciplinary research; addresses complex societal challenges at a systemic level; co-)principal investigator in a dozen national and European projects addressing bottom-up participatory innovation; the European Network of Living Labs is tangible outcome of her pioneering Living Labs research; led impact evaluation of the first Dutch living lab “Kenniswijk” and recently the H2020 DESIGNSCAPES capacity building program to scale design-enabled urban innovation.²⁹

CvO-Clarine van Oel: environmental psychologist and associate professor in the Department of Architecture & the Built Environment at TU Delft; 1989-2006 conducted epidemiological and user-driven research in Healthy Environments (TNO Bouw in Delft), public health (Amsterdam Medical Center), and psychiatry (Utrecht Medical Center); since 2006, investigates user-preferences for design solutions using visual display methods; granted together with Brazil, UK and Germany the NWO project uVITAL project as part of the Trans-Atlantic-Platform “Social Innovation” (2019). Together with prof. E. Eisemann (co-applicant) the VR-Renovate project (2020).³⁰⁻³²

LO-Lieke Oldenhof: expertise on decentralization of care (Veni, NWO; Erasmus Research Excellence Initiative -The State of Local Welfare), participation in urban labs & social resilience (National Science Agenda, JOIN consortium), healthy neighbourhoods & networks (ZonMw evaluation of local neighbourhood networks & VWS program Neighbourhood-based working), the digital welfare state & use of big data (multi-disciplinary research team ESHPM & supervision Phd students), narrative methods, action research and multi-sited ethnography; co-initiated CARE Lab Rotterdam to improve the health of people with multiple problems (partners: the municipality of Rotterdam, ESHPM and Higher Education Institutes); supervised 7 PhD students, 40+ Master and Bachelor Thesis Projects at ESHPM.³³⁻³⁵

RST-Régine Steegers-Theunissen: periconception epidemiologist and medical doctor at Erasmus MC; (inter)national pioneer in periconception research and eHealth development, with interests related to the Developmental Origins of Health and Disease paradigm and focus on the periconception period as “earliest window of the life course” involving the role of lifestyle in the complex pathophysiology of birth defects, low birth weight and subfertility; PI of Rotterdam Periconception cohort (Predict study, >2009) embedded in patient care with focus on periconception parental health determinants, epigenetics, and periconception- and long term health outcomes in mother and child; initiated and implemented a successful preconception outpatient clinic for tailored nutrition and lifestyle counselling (2007) and developed web-based eHealth coaching programmes on the mobile phone, eg. Smarter Pregnancy; partner in the EU projects LIFECYCLE (Horizon2020), DohArt-net (ITN), and JPI Precise (ERA).³⁷⁻⁵⁴

JvS-Jacco van Sterkenburg: Endowed Professor of 'Race', Inclusion and Communication in the Department of Media & Communication at Erasmus University; research focuses on issues related to media, inclusion and diversity, mainly in relation to sport and physical activity; currently

coordinates a sub-project within a ZON-MW funded project on health and sport experiences amongst urban residents in disadvantaged neighborhoods in Rotterdam (with Radboud Universiteit).

VV-Valentijn Visch: associate professor at the faculty of Industrial Design Engineering, TUDelft; research is focused on Design for Health Motivation and is creative, user-experience centered, persuasive game design inspired and typically involves multidisciplinary teams like creative industry, healthcare institutions, patients, medical practitioners, and researchers from various domains; chairs the eHealth Design lab at the TU Delft and has a background in literature (MA), art theory (MA), animation (NIAf), cognitive film studies (PhD) and emotion research (PD); coordinates NWO project on storytelling as a persuasive design tool for eHealth, an interdisciplinary project on Digital Twins and a ZonMw project on longterm behavioral change (Enduring Rewards); is a scientific lead in the Medical Delta 'eHealth and Selfmanagement' research group and in the Convergence research theme 'Technology supported transitions in Health(care)'.⁵⁵⁻⁵⁸

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Principle actors and tasks are distributed: IM with the task of defining our vision, and LO of relating it to contemporary context (Group A, with SC, VV, NV, FM, and Group B, DA). WJ (with IM, DA) and other Group B investigators (incl. VV, CvO, RST, JvS) work together in translating HN theory to living lab criteria; the postdoc works toward matching these with active technology interventions. JK will ensure data collection informs further development of principles (Groups B & C, with CvO, VV, WJ), and SC addresses the implications for HN in terms of generalization, extracting further potential, and ensuring justice in process development (Group C, with IM, BB, FM, LO).

Groups A, B and C represent key focal points for the project, but between them co-ordinated efforts also produce results and allow movement between aims. A and B together work toward co-creation of living lab contexts for pilots; B and C together look to the results of pilots broadly, to measure effects on motivation and attitudes of endusers and other stakeholders. Further collaboration between C and A develops principles as well as evaluating them in context (relevance and effectiveness in terms of current policy and practice). Together, they further look explicitly for untapped potential in current systems that has been identified during the research, and bring these reflections back into A for further work on the HN concept.

	A- Concept & Vision	B- Health Technology Pilots	C- Impact Assessment & ToC
Objectives (see §3b)	HN framework	Co-creation of living labs; Applying HN principles	Ensuring uptake; Iterative evaluation of HN

Methods (see §3c)	Ethnography, co-creation and participatory methods (focus groups, walk-along interviews, digital ethnography/phenotyping)	Quasi-experimental contexts, eg. living labs and digital spaces, developed via co-creation and participatory methods; process evaluation via surveys and/or experimental approaches	Ethical analysis, evaluation of health outcomes and attitudes toward technology (assessing motivation and engagement in end-users & stakeholder networks);
Outcomes	Multi-disciplinary, multi-layered, multi-scalar framework for HN;	Principles for tailoring technologies (in-context, specific interventions)	Generalized principles for tailoring technology (integration of results from pilots & Group B) + refinement and enhancement of HN (with Group A); progressive Theory of Change
Funded staff (see §4d)	2+ PhD	1+ PhD (+ postdoc assistance)	1+ PHD + 1+ postdoc

4.c Involvement of partners

Several members of the consortium already work together closely with the municipality of Rotterdam and are well-connected with partners in the ecosystem, including health and care providers and local neighbourhood partners and communities through existing networks and academic workplaces that focus on healthy neighbourhoods, such as CEPHIR, ST-RAW, Kenniswerkplaats Leefbare Wijken, Regionaal Consortium Zwangerschap en Geboorte Zuid-West Nederland, CARE LAB Rotterdam, Enduring Rewards (Departments of Sports).

New (already initiated) partnerships will enable comparisons and validate the results from our work in Rotterdam by collaborating with Medical Delta, optimising current connections between team members and that initiative. Living Labs for uVITAL and VR-Renovate are also developed in collaboration with the VR-Zone (TU Delft, New Media Center; and prof. Eisemann - Computer Visualisation Group EWI), and we are in discussion with the Province Zuid-Holland (Ronald Dijkgraaf) for a living lab within Rotterdam (CvO).

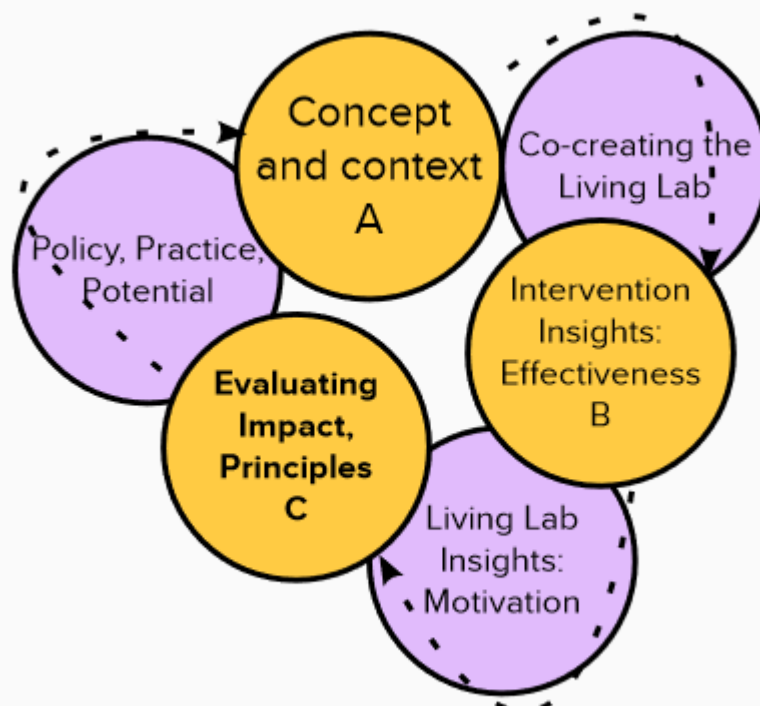
4.d Governance, organization and collaboration strategy of the consortium

PhDs and a postdoc are hired to assist with fundamental research and organizing collaborative efforts. Exchanges between these and PhDs (each supervised by committee members from each of the three institutions) ensure an ongoing, inter-institutional academic community. In particular the postdoc hire is tasked with generating collaboration opportunities for transdisciplinary work

and optimizing resources already present among us toward a stronger network and effective dissemination of results. As both HN and project success require a great deal of transdisciplinary and Interdisciplinary cooperation and translational work, this is both an outcome and part of the research process.

Communication within the flagship is a challenge. Organization under three groups (see §4B table) creates sub-networks between institutions and among researchers. These groups further create opportunities for smaller, regular gatherings under their themes, to share and summarise results for communication between groups. Annual meetings will share results, as well as taking up training and co-writing opportunities created by the Main Lead (SC) and postdocs. Each group contains a mix of senior researchers with considerable experience in achieving outcomes through interdisciplinary collaboration, and junior collaborators with particular expertise to contribute, so that the flagship will create a hub of interdisciplinary and transdisciplinary research toward a common HN approach.

This graphic represents the movement between the groups, illustrating how the flagship is organized under the principles of progressive iteration, inclusivity and bottom-up development of principles for theory, practice, and design:



Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

Pluriannual plan provided.

II Letters of intent and letters of commitment

Letter of intent from City of Rotterdam provided.

III Letter of support from department head of the main lead

Letter of support for Samantha Copeland provided.

IV References

1. Adam W. A. Geraghty, Leandro D. Torres, Yan Leykin, Eliseo J. Pérez-Stable, Ricardo F. Muñoz. 2013. Understanding attrition from international internet health interventions: a step towards global eHealth. *Health Promotion International*, Volume 28, Issue 3, September 2013, Pages 442–452, <https://doi.org/10.1093/heapro/das029>
2. Wendy Brouwer, Anke Oenema, Hein Raat, Rik Crutzen, Jascha de Nooijer, Nanne K. de Vries, Johannes Brug. 2010. Characteristics of visitors and revisitors to an Internet-delivered computer-tailored lifestyle intervention implemented for use by the general public. *Health Education Research*, Volume 25, Issue 4, August 2010, Pages 585–595, <https://doi.org/10.1093/her/cyp063>
3. Fabienne Reiners, Janienke Sturm, Lisette J.W. Bouw and Eveline J.M. Wouters. 2019. Sociodemographic Factors Influencing the Use of eHealth in People with Chronic Diseases. *Int. J. Environ. Res. Public Health* 16(4), 645; <https://doi.org/10.3390/ijerph16040645>
4. David Hunter, 2012. Promoting health equity (Editorial) *BMJ*; 345 doi: <https://doi.org/10.1136/bmj.e4881>
5. <https://www.rotterdam.nl/wonen-leven/wijkprogrammering/>
6. Ingrid Mulder, 2018. Co-creative partnerships as catalysts for social change. *Strategic Design Research Journal*, 11(3), 178-185.
7. Designing for healthier citizens (Hospital Hero and Healthy Eating/Parenting). <https://designscapes.eu/2021/09/15/thursday-16-september-2021designing-for-healthier-citizens-hospital-hero-and-healthy-eating-parenting/>
8. Fierloos, I. N., Windhorst, D. A., Fang, Y., Mao, Y., Crone, M. R., Hosman, C. M. H., Jansen, W., Raat, H. (2022). Factors associated with media use for parenting information: A cross-sectional study among parents of children aged 0-8 years. *Nurs Open*, 9(1), 446-457. doi:10.1002/nop2.1084
9. Mieloo, C. L., van der Ende, J., van Zijl, A. L., Schuring, M., Steijn, B., & Jansen, W. (2022). Changes in youth care use after the implementation of community-based support teams: repeated measurement study using registry data and data on team characteristics. *BMJ Open*, 12(2), e048933. doi:10.1136/bmjopen-2021-048933
10. Smit, M. S., Raat, H., Molenberg, F., Wolfers, M. E. G., Bannink, R., & Jansen, W. (2021). Study protocol for the evaluation of long-term effects of the school-based obesity

- prevention program Lekker Fit! ('enjoy being fit'): a retrospective, controlled design. *BMJ Open*, 11(8), e046940. doi:10.1136/bmjopen-2020-046940
11. Yang-Huang, J., van Grieken, A., You, Y., Jaddoe, V. W. V., Steegers, E. A., Duijts, L., Boelens, M., Jansen, W., Raat, H. (2021). Changes in Family Poverty Status and Child Health. *Pediatrics*, 147(4). doi:10.1542/peds.2020-016717
 12. Boelens, M., Raat, H., Yang-Huang, J., Schouten, G. M., van Grieken, A., & Jansen, W. (2020). Independent contributions of family and neighbourhood indicators of socioeconomic status and migrant status to risk of mental health problems in 4-12-year-old children. *SSM Popul Health*, 12, 100675. doi:10.1016/j.ssmph.2020.100675
 13. Boelens, M., Windhorst, D. A., Jonkman, H., Hosman, C. M. H., Raat, H., & Jansen, W. (2019). Evaluation of the promising neighbourhoods community program to reduce health inequalities in youth: a protocol of a mixed-methods study. *BMC Public Health*, 19(1), 555. doi:10.1186/s12889-019-6901-3
 14. Jansen, W., Kruitwagen-van de Gaar, V. M. J., Blanchette, L. M. G., & Raat, H. (2017). Sociale marketing inzetten voor gezond gedrag. *Tijdschrift voor gezondheidswetenschappen*, 95(7), 307-316. doi:10.1007/s12508-017-0093-y
 15. Wang, L., van de Gaar, V. M., Jansen, W., Mieloo, C. L., van Grieken, A., & Raat, H. (2017). Feeding styles, parenting styles and snacking behaviour in children attending primary schools in multiethnic neighbourhoods: a cross-sectional study. *BMJ Open*, 7(7), e015495. doi:10.1136/bmjopen-2016-015495
 16. Wijtzes, A. I., van de Gaar, V. M., van Grieken, A., de Kroon, M. L., Mackenbach, J. P., van Lenthe, F. J., Jansen, W., Raat, H. (2017). Effectiveness of interventions to improve lifestyle behaviors among socially disadvantaged children in Europe. *Eur J Public Health*, 27(2), 240-247. doi:10.1093/eurpub/ckw136
 17. Jansen, W., Mieloo, C. L., Anschutz, J., & de Zwart, O. (2015). [Gap between the use of and need for youth care: research in Rotterdam neighbourhoods]. *Ned Tijdschr Geneesk*, 159, A7664.
 18. Wijtzes, A. I., Jansen, W., Bouthoorn, S. H., Pot, N., Hofman, A., Jaddoe, V. W., & Raat, H. (2014). Social inequalities in young children's sports participation and outdoor play. *Int J Behav Nutr Phys Act*, 11, 155. doi:10.1186/s12966-014-0155-3
 19. Jansen, W., Borsboom, G., Meima, A., Zwanenburg, E. J., Mackenbach, J. P., Raat, H., & Brug, J. (2011). Effectiveness of a primary school-based intervention to reduce overweight. *Int J Pediatr Obes*, 6(2-2), e70-77. doi:10.3109/17477166.2011.575151
 20. Jansen, W., Mackenbach, J. P., Joosten-van Zwanenburg, E., & Brug, J. (2010). Weight status, energy-balance behaviours and intentions in 9-12-year-old inner-city children. *J Hum Nutr Diet*, 23(1), 85-96. doi:10.1111/j.1365-277X.2009.01027.x
 21. Jansen, W., & Hazebroek-Kampschreur, A. A. (1997). Differences in height and weight between children living in neighbourhoods of different socioeconomic status. *Acta Paediatr*, 86(2), 224-225.
 22. Kneer, J., van Eldik, A., Jansz, J., Eischeid, S., & Usta, M. (2019). With a little help from my friends: Peer coaching for refugee adolescents and the role of social media. *Media and Communication*, 7(2), 264- 274.
 23. Kneer, J. & Jansz, J. (2017, 20th December). The fogland game. YouTube. <https://www.youtube.com/watch?v=1p0kmh04B2Q>
 24. Van Eldik, A., Kneer, J., & Jansz, J. (2019). Urban & Online: Social media use amongst adolescents and sense of belonging to a super-diverse city. *Media and Communication*, 7(2), 242-253.

25. Van Eldik, A. K., Kneer, J., Lutkenhaus, R. O., & Jansz, J. (2019). Urban influencers: An analysis of urban identity in YouTube content of local social media influencers in a super-diverse city. *Frontiers in psychology*, 2876.
26. Mölenberg FJM, Mackenbach JD, Poelman MP, Santos S, Burdorf A, van Lenthe FJ. (2021). Socioeconomic inequalities in the food environment and body composition among school-aged children: a fixed-effects analysis. *International Journal of Obesity*, 45 (12): 2554-2561.
27. Mölenberg FJM, de Vries C, Burdorf A, van Lenthe FJ. (2021) A framework for exploring non-response patterns over time in health surveys. *BMC Medical Research Methodology*. 18;21(1):37.
28. Mölenberg FJM, Noordzij JM, Burdorf A, van Lenthe FJ. (2019) New physical activity spaces in deprived neighborhoods: Does it change outdoor play and sedentary behavior? A natural experiment. *Health Place*, 58:102151.
29. Mulder, I. (2018). Co-creative partnerships as catalysts for social change. *Strategic Design Research Journal*, 11(3), 178-185.
30. Van Oel, C. J., Mlihi, M., & Freeke, A. (2021). Design models for single patient rooms tested for patient preferences. *HERD: Health Environments Research & Design Journal*, 14(1), 31-46.
31. Bentinck, S. A., Van Oel, C. J., & Van Dorst, M. J. (2020). Perception of privacy in a university building: The transparency paradox. *Frontiers of Architectural Research*, 9(3), 579-587.
32. Papadonikolaki, E., Van Oel, C.J., & Kagioglou, M. (2019). Organising and Managing boundaries: A structurational view of collaboration with Building Information Modelling (BIM). *International journal of project management*, 37(3), 378-394.
33. L.E. Oldenhof, S. Rahmawan- Huizenga, H. van de Bovenkamp and R. Bal (2020). The governance challenge of Urban Living Labs: using the 'in-between' liminal space to create livable cities. In: C. van Montfort and A. Michels ed. *Partnerships for livable cities*. Routledge.
34. L.E. Oldenhof, J. Postma and R. Bal (2016). Re-placing care. *Governing care through spatial arrangements*. In: ed. E. Ferlie, K. Montgomery and A.R. Pedersen. *Oxford Handbook of Health Care Management*. Oxford: Oxford University Press.
35. L.E. Oldenhof (2013). Wijkgerichte samenwerking dankzij grenzenwerk. *Gerón. Tijdschrift over ouder worden & maatschappij* 15 (4): 43-46.
36. Rani Lill Anjum, Samantha Copeland, Elena Rocca. 2020. *Rethinking Causality, Complexity and Evidence for the Unique Patient*. Springer <https://link.springer.com/book/10.1007/978-3-030-41239-5>
37. Ng KYB, Steegers-Theunissen RPM, Willemsen SP, Wellstead S, Cheong Y, Macklon N. Smartphone based lifestyle coaching programme 'Smarter Pregnancy' modifies behaviours in women with subfertility or recurrent miscarriage who are planning a pregnancy: A Randomised Controlled Trial *Reproductive BioMedicine Online* ISSN 1472-6483. <https://doi.org/10.1016/j.rbmo.2021.04.003>
38. Steegers-Theunissen RPM, Hoek A, Groen H, Bos A, Van den Dool G, Schoonenberg M, Smeenk J, Creutzberg E, Vecht L, Starmans L, Laven J. Pre-conception interventions for subfertile couples undergoing assisted reproductive technology treatment: modeling analysis. *JMIR Mhealth Uhealth* 2020; 8 (11): e19570 doi:10.2196/19570 <https://mhealth.jmir.org/2020/11/e19570>
39. Van der Windt M, Van der Kleij R, Snoek K, Willemsen S, Dijkgraaf R, Laven J, Schoenmakers S, Steegers-Theunissen R. Impact of a blended periconception lifestyle care approach on lifestyle behaviors: before-and-after study. *J Med Internet Res* 2020; 22 (9): e19378 doi:10.2196/19378 <https://www.jmir.org/2020/9/e19378/>

40. Oostingh EC, Koster MPH, Van Dijk MR, Willemsen SP, Broekmans FJM, Hoek A, Goddijn M, Klijn NF, van Santbrink EJP, Steegers EAP, Laven JSE, Steegers-Theunissen RPM. First effective mHealth nutrition and lifestyle coaching program for subfertile couples undergoing in vitro fertilization treatment: a single-blinded multicenter randomized controlled trial. *Fertil Steril*. 2020 Jul 30; S0015-0282 (20) 30418-0. doi:10.1016/j.fertnstert.2020.04.051 <https://pubmed.ncbi.nlm.nih.gov/32741619/>
41. Hoek J, Schoenmakers S, Baart EB, Koster MPH, Willemsen SP, Van Marion ES, Steegers EAP, Laven JSE, Steegers-Theunissen RPM. Preconceptional maternal vegetable intake and paternal smoking are associated with pre-implantation embryo quality. *Reproductive Sciences*, 15 June 2020, <https://doi.org/10.1007/s43032-020-00220-8>
42. Laureij LT, Breunis LJ, Steegers-Theunissen RPM, Rosman AN. Identifying the needs for a web-based postpartum platform among parents of newborns and health care professionals: qualitative focus group study. *JMIR Form Res* 2020, vol. 4, iss. 5, e16202 <https://formative.jmir.org/2020/5/e16202>
43. Van Dijk MR, Koster MPH, Oostingh EC, Willemsen SP, Steegers EAP, Steegers-Theunissen RPM. A Mobile App Lifestyle Intervention to Improve Healthy Nutrition in Women Before and During Early Pregnancy: Single-Center Randomized Controlled Trial. *J Med Internet Res* 2020; 22 (5): e15773, doi: 10.2196/15773 <https://www.jmir.org/2020/5/e15773>
44. Oostingh EC, Ophuis RH, Koster MPH, Polinder S, Lingsma HF, Laven JSE, Steegers-Theunissen RPM. Mobile health coaching on nutrition and lifestyle behaviors for subfertile couples using the Smarter Pregnancy Program: model-based cost-effectiveness analysis. *JMIR Mhealth Uhealth* 2019 vol. 7 iss.10 e13935 1-9 <https://mhealth.jmir.org/2019/10/e13935/>
45. Gootjes DV, Van Dijk MR, Koster MPH, Willemsen SP, Steegers EAP, Steegers-Theunissen RPM. Neighborhood deprivation and the effectiveness of mobile health coaching to improve periconceptional nutrition and lifestyle in women: survey in a large urban municipality in the Netherlands. *JMIR Mhealth Uhealth* 2019, 7: e11664. <https://mhealth.jmir.org/2019/4/e11664/>
46. Barker M, Dombrowski SU, Colbourn T, Fall CHD, Kriznik NM, Lawrence WT, Norris SA, Ngaiza G, Patel D, Skordis-Worrall J, Sniehotta FF, Steegers-Theunissen RPM, Vogel C, Woods-Townsend K, Stephenson J. Intervention strategies to improve nutrition and health behaviors before conception. *The Lancet* 2018, April 16, [https://dx.doi.org/10.1016/S0140-6736\(18\)30313-1](https://dx.doi.org/10.1016/S0140-6736(18)30313-1)
47. Overdijkink SB, Velu AV, Rosman AN, Van Beukering MDM, Kok M, Steegers-Theunissen RPM. The usability and effectiveness of mobile health technology-based lifestyle and medical intervention apps supporting health care during pregnancy: systematic review. *JMIR Human Factors* 2018 24 April, Vol 6, No 4 <https://dx.doi.org/10.2196/mhealth.8834>
48. Van Dijk MR, Oostingh EC, Koster MP, Willemsen SP, Laven JS, Steegers-Theunissen RP. The use of the mHealth program Smarter Pregnancy in preconception care: rationale, study design and data collection of a randomized controlled trial. *BMC Pregnancy Childbirth*. 2017 Jan 26; 17 (1): 46. doi: 10.1186/s12884-017-1228-5. <https://bmcpregnancychildbirth.biomedcentral.com/articles/10.1186/s12884-017-1228-5>
49. Van Dijk MR, Koster MPH, Rosman AN, Steegers-Theunissen RPM. Opportunities of mHealth in Preconception Care: Preferences and Experiences of Patients and Health Care Providers and Other Involved Professionals. *JMIR Mhealth Uhealth* 2017; 5 (8): e123. <https://mhealth.jmir.org/2017/8/e123/>
50. Van Dijk MR, Koster MPH, Willemsen SP, Huijgen NA, Laven JSE, Steegers-Theunissen RPM. Healthy preconception nutrition and lifestyle using personalized mobile health coaching is

- associated with enhanced pregnancy chance. *Reprod Biomed Online* 2017 Oct; 35 (4): 453–460. doi: 10.1016/j.rbmo.2017.06.014. Epub 2017 Jun 27. PMID: 28688924.
<https://www.sciencedirect.com/science/article/pii/S1472648317302675?via%3Dihub>
51. Steegers-Theunissen RPM. Periconception mHealth platform for prevention of placental-related outcomes and non-communicable diseases. *Placenta* 2017 Apr 23. pii: S0143-4004 (17) 30239-4. doi: 10.1016/j.placenta.2017.04.017.
<https://www.sciencedirect.com/science/article/pii/S0143400417302394?via%3Dihub>
 52. Van Dijk MR, Huijgen NA, Willemsen SP, Laven JSE, Steegers EAP, Steegers-Theunissen RPM. Impact of an mHealth Platform for Pregnancy on Nutrition and Lifestyle of the Reproductive Population: A Survey. *JMIR mHealth uHealth* 2016 May 27; 4 (2): e53.
<https://www.ncbi.nlm.nih.gov/pmc/articles>
 53. Steegers EAP, Barker ME, Steegers-Theunissen RPM, Williams MA. Societal Valorisation of New Knowledge to Improve Perinatal Health: Time to Act. *Paediatr Perinat Epidemiol*. 2016 Mar; 30 (2): 201-4. <https://onlinelibrary.wiley.com/doi/full/10.1111/ppe.12275>
 54. Steegers-Theunissen RPM, Steegers EA. Embryonic health: new insights, mHealth and personalised patient care. *Reprod Fertil Dev*. 2015; 27 (4): 712-16.
<https://www.publish.csiro.au/rd/RD14386>
 55. I Al-Dhahir, T Reijnders, RJG van den Berg-Emons, V Visch, et al. Possibilities and challenges in ehealth lifestyle interventions for people with low socioeconomic status: a scoping review.
INTERNATIONAL JOURNAL OF BEHAVIORAL MEDICINE 28 (SUPPL 1), S141-S141
 56. JS Faber, I Al-Dhahir, T Reijnders, NH Chavannes, AWM Evers, JJ Kraal, et al. Attitudes Toward Health, Healthcare, and eHealth of People With a Low Socioeconomic Status: A Community-Based Participatory Approach. *Frontiers in Digital Health*, 68
 57. E Knutz, T Markussen, V Visch, E Tan. Family narratives and identity navigation in social design: a method of analysis. *Journal of Design Research* 18 (5-6), 388-409
 58. P Siriaraya, V Visch, A Vermeeren, M Bas. A cookbook method for Persuasive Game Design. *International Journal of Serious Games* 5 (1), 37-71
 59. Natasha Azzopardi-Muscat, Kristine Sørensen, Towards an equitable digital public health era: promoting equity through a health literacy perspective, *European Journal of Public Health*, Volume 29, Issue Supplement_3, October 2019, Pages 13–17,
<https://doi.org/10.1093/eurpub/ckz166>
 60. Kaharevic, A., & Skill, K. (2021). Digital Citizenship in a Swedish Marginalised Neighbourhood: Different attitudes to and experiences of digital inclusion and eHealth. *JeDEM - EJournal of EDemocracy and Open Government*, 13(1). <https://doi.org/10.29379/jedem.v13i1.637>
 61. Liaw, S.-T., & De Lusignan, S. (2019). An "integrated health neighbourhood" framework to optimise the use of EHR data. *Journal of Innovation in Health Informatics*, 23(3), 826.
 62. Puerari, E.; De Koning, J.I.J.C.; Von Wirth, T.; Karré, P.M.; Mulder, I.J.; Loorbach, D.A. Co-Creation Dynamics in Urban Living Labs. *Sustainability* 2018, 10, 1893.
<https://doi.org/10.3390/su10061893>
 63. Breuer, E., De Silva, M., & Lund, C. (2018). Theory of change for complex mental health interventions: 10 lessons from the programme for improving mental healthcare. *Global Mental Health*, 5, E24. doi:10.1017/gmh.2018.13
 64. Pennie Frow, Janet R. McColl-Kennedy, Adrian Payne, Co-creation practices: Their role in shaping a health care ecosystem, *Industrial Marketing Management*, Volume 56, 2016, Pages 24-39, ISSN 0019-8501, <https://doi.org/10.1016/j.indmarman.2016.03.007>.

Budget formulier: Convergentie Flagship Call

Flagship:	Tailoring Technologies for a Healthy Neighbourhood
Flagship acronym:	TTHN
Naam Penvoerder:	Samenita Copeland
Startdatum en looptijd van het project:	01-09-2022, 5 years
Totaal project budget:	€ 2.000.000,- by Convergence (€ 400.000,- per year) and € 2.000.000,- by matched funding

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner		
	Partner die de kosten maakt	Functie	Aantal Re	Startdatum	Einddatum	Inschaling	Trede	Maandbedrag	Kosten	in cash		in kind		in cash		in kind		in cash		in kind		Totaal		Totaal per partner	
Onderzoeksorganisaties																									
	TU Delft	OIO/PHD	2.00	01/11/2022	31/10/2026	UNI - p	3	€ 3.122	€ 449.568	€ -	€ -	€ 37.464	€ -	€ 112.392	€ -	€ 112.392	€ -	€ 112.392	€ -	€ 74.928	€ -	€ 449.568	€ 449.568		
	TU Delft	postdoc	0.10	01/09/2022	31/08/2027	UNI - 10	10	€ 4.222	€ 37.998	€ -	€ -	€ -	€ 4.222	€ 12.666	€ 12.666	€ -	€ 8.444	€ -	€ 37.998	€ -	€ 37.998	€ 37.998			
	TU Delft	student assistant	1.00	01/09/2022	31/08/2027	UNI - SA	2	€ 2.310	€ 207.900	€ 13.860	€ -	€ 41.580	€ 41.580	€ 207.900	€ 41.580	€ -	€ 27.720	€ -	€ 207.900	€ -	€ 207.900	€ 207.900			
	TU Delft	postdoc	0.20	01/01/2023	31/12/2027	UNI - 10	10	€ 4.222	€ 75.996	€ 5.066	€ -	€ 15.199	€ 15.199	€ 75.996	€ 15.199	€ -	€ 10.133	€ -	€ 75.996	€ -	€ 75.996	€ 75.996			
	EUR	OIO/PHD	1.00	01/11/2022	31/10/2026	UNI - p	3	€ 3.122	€ 224.784	€ -	€ 18.732	€ 56.196	€ 56.196	€ 56.196	€ 56.196	€ -	€ -	€ 224.784	€ -	€ 224.784	€ 224.784				
	EUR	postdoc	0.50	01/01/2023	31/12/2027	UNI - 10	3	€ 2.282	€ 147.690	€ 12.308	€ -	€ 36.923	€ 36.923	€ 147.690	€ 36.923	€ -	€ -	€ 147.690	€ -	€ 147.690	€ 147.690				
	ErasmusMC	OIO/PHD	1.00	01/11/2022	31/10/2026	UMC - 10	7	€ 4.080	€ 293.760	€ 24.480	€ -	€ 73.440	€ 73.440	€ 293.760	€ 73.440	€ -	€ -	€ 293.760	€ -	€ 293.760	€ 293.760				
	ErasmusMC	postdoc	0.50	01/01/2023	31/12/2027	UMC - 10	12	€ 4.800	€ 216.000	€ -	€ -	€ 54.000	€ 54.000	€ 216.000	€ 54.000	€ -	€ -	€ 216.000	€ -	€ 216.000	€ 216.000				
Ondernemingen																									
	Onderneming 1							€ -	€ 0											€ 0	€ 0	€ -	-		
	Onderneming 2							€ -	€ 0											€ 0	€ 0	€ -	-		
	Onderneming 3							€ -	€ 0											€ 0	€ 0	€ -	-		
	Onderneming 4							€ -	€ 0											€ 0	€ 0	€ -	-		
Overige (private) partners Matching																									
	ZonMW (VV - Healthy Storytelling -2022)	Junior Researcher	1.00	01/09/2022	31/08/2023	UNI - 8	1	€ 2.846	€ 51.228	€ 17.076	€ -	€ 34.152	€ -	€ 227.988	€ -	€ 227.988	€ -	€ 151.992	€ -	€ 51.228	€ -	€ 51.228	€ 51.228		
	NWO (VV - Enduring Rewards -2024)	OIO/PHD	3.00	01/09/2022	31/08/2026	UNI - 10	10	€ 4.222	€ 911.952	€ -	€ 75.996	€ 227.988	€ 227.988	€ -	€ 227.988	€ -	€ -	€ 911.952	€ -	€ 911.952	€ -	€ 911.952	€ 911.952		
	Benefit4M (Medical Delta, VV - eHealth design)	postdoc	0.20	01/09/2022	31/12/2023	UNI - 10	10	€ 4.222	€ 20.266	€ 5.066	€ -	€ 15.199	€ 15.199	€ 20.266	€ 15.199	€ -	€ -	€ 20.266	€ -	€ 20.266	€ -	€ 20.266	€ 20.266		
	TAP SI (CoV - uVital - 2023)	Junior Researcher	0.80	01/09/2022	31/12/2023	UNI - 9	1	€ 3.122	€ 59.942	€ 19.981	€ -	€ 39.962	€ -	€ 59.942	€ -	€ 59.942	€ -	€ -	€ 59.942	€ -	€ 59.942	€ -	€ 59.942	€ 59.942	
	NWO (CoV - VR-Renovate -2025)	OIO/PHD	0.53	01/09/2022	31/08/2024	UNI - p	3	€ 3.122	€ 59.568	€ 9.928	€ -	€ 29.784	€ 19.856	€ 59.568	€ 19.856	€ -	€ -	€ 59.568	€ -	€ 59.568	€ -	€ 59.568	€ 59.568		
	EU Horizon (VV Obesity Prevention 2023-27)	OIO/PHD	2.00	01/09/2023	31/08/2027	UNI - p	3	€ 3.122	€ 449.568	€ -	€ 37.464	€ 112.392	€ 112.392	€ 112.392	€ 112.392	€ -	€ 74.928	€ -	€ 449.568	€ -	€ 449.568	€ 449.568			
	Horizon NEB (IM 2022-25)	postdoc	0.20	01/01/2023	31/12/2025	UMC - 10	10	€ 4.506	€ 48.665	€ -	€ 16.222	€ 16.222	€ 16.222	€ 16.222	€ -	€ -	€ 48.665	€ -	€ 48.665	€ -	€ 48.665	€ 48.665			
	NWO-KIC (CoV Learning Community 2023-28)	postdoc	0.50	01/09/2025	31/08/2027	UMC - 10	10	€ 4.506	€ 81.108	€ -	€ 27.036	€ 27.036	€ 27.036	€ 27.036	€ -	€ 27.036	€ 27.036	€ -	€ 81.108	€ -	€ 81.108	€ 81.108			
	NWO-KIC (VV, SC, IM, LO, WJ, Medical Delta 2023-28)	postdoc	1.00	01/09/2022	31/08/2027	UNI - 10	10	€ 4.222	€ 379.980	€ 25.332	€ -	€ 75.996	€ 75.996	€ 379.980	€ 75.996	€ -	€ 50.664	€ -	€ 379.980	€ -	€ 379.980	€ 379.980			
	NWO-KIC lifestyle and living envr.	postdoc	2.00	01/09/2022	31/08/2026	UNI - 12	5	€ 5.294	€ 762.336	€ 63.528	€ -	€ 190.584	€ 190.584	€ 190.584	€ 190.584	€ -	€ 127.056	€ -	€ 762.336	€ -	€ 762.336	€ 762.336			
Totaal									€ 4.478.309															Totaal loonkosten	€ 4.478.309

2.	Kosten van materialen en hulpmiddelen:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner		
	Partner die de kosten maakt	Materiaal / hulpmiddel						Kosten	in cash		in kind		in cash		in kind		in cash		in kind		Totaal		Totaal per partner		
Onderzoeksorganisaties																									
	EUR	Huawei Media Pad 16 GB x12						€ 1.400	€ 1.400	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ 1.400	€ 1.400		
	TU Delft	VR equipment						€ 4.000	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ -	€ 4.000	€ 4.000		
	ErasmusMC	user studies						€ 15.000	€ 1.000	€ -	€ 3.000	€ -	€ 3.000	€ -	€ 3.000	€ -	€ 3.000	€ -	€ 15.000	€ -	€ 15.000	€ 15.000			
	TU Delft	licences						€ 50.000			€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 50.000	€ 50.000			
Ondernemingen																									
	Onderneming 1																				€ 0	€ 0	€ 0,00		
	Onderneming 2																				€ 0	€ 0	€ 0,00		
	Onderneming 3																				€ 0	€ 0	€ 0,00		
	Onderneming 4																				€ 0	€ 0	€ 0,00		
Overige (private) partners																									
	Hanarth fund																				€ 0	€ 0	€ 0,00		
	Naam 2																				€ 0	€ 0	€ 0,00		
	Naam 3																				€ 0	€ 0	€ 0,00		
	Naam 4																				€ 0	€ 0	€ 0,00		
Totaal									€ 70.400															Totaal kosten materialen en hulpmiddelen	€ 70.400

Kosten van gebruik van machines en apparatuur							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner		
Partner die de kosten maakt	Machine / apparatuur			Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties																			€ 0	€ 0	€ 0,00		
...																			€ 0	€ 0			
...																			€ 0	€ 0			
...																			€ 0	€ 0			
Ondernemingen																				€ 0	€ 0	€ 0,00	
Onderneming 1																			€ 0	€ 0			
Onderneming 2																			€ 0	€ 0			
Onderneming 3																			€ 0	€ 0			
Onderneming 4																				€ 0	€ 0	€ 0,00	
Overige (private) partners																				€ 0	€ 0		
Naam 1																				€ 0	€ 0		
Naam 2																				€ 0	€ 0		
Naam 3																					€ 0	€ 0	€ 0,00
Naam 4																				€ 0	€ 0		
Totaal							€ 0															Totaal kosten machines en apparatuur	€ 0

Aan derden verschuldigde kosten				2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt		Naam derde, omschrijving kosten		Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																		
...	hiring third party (programmers etc.)		€ 248.404					€ 49.681		€ 49.681		€ 49.681		€ 49.681		€ 49.681		€ 248.404
...																		€ 0.00
...																		€ 0.00
...																		€ 0.00
Ondernemingen																		
Onderneming 1																		€ 0.00
Onderneming 2																		€ 0.00
Onderneming 3																		€ 0.00
Onderneming 4																		€ 0.00
Overige (private) partners																		
Naam 1																		€ 0.00
Naam 2																		€ 0.00
Naam 3																		€ 0.00
Naam 4																		€ 0.00
Totaal				€ 248.404														€ 248.404
Totaal kosten derden																		€ 248.404

Publicatie, reis- en verblijfkosten					2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
Partner die de kosten maakt	Naam derde, omschrijving kosten				Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																				
ErasmusMC	Travel & conferences (€ 1.000,- per scientific FTE per year)				€ 3.000	€ -	€ -	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ -	3000	€ 0	€ 3.000,00
TU Delft	Travel & conferences (€ 1.000,- per scientific FTE per year)				€ 12.000	€ -	€ -	€ -	€ 4.000,00	€ -	€ 4.000,00	€ -	€ 4.000,00	€ -	€ 4.000,00	€ -	€ -	€ 12.000	€ 0	€ 12.000,00
EUR	Travel & conferences (€ 1.000,- per scientific FTE per year)				€ 3.000	€ -	€ -	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ 1.000,00	€ -	€ -	€ 3.000	€ 0	€ 3.000,00
ErasmusMC	Open access publications				€ 1.500	€ -	€ -	€ -	€ 500,00	€ -	€ 500,00	€ -	€ 500,00	€ -	€ 500,00	€ -	€ -	€ 1.500	€ 0	€ 1.500,00
TU Delft	Open access publications				€ 6.500	€ -	€ -	€ -	€ 2.166,67	€ -	€ 2.166,67	€ -	€ 2.166,67	€ -	€ 2.166,67	€ -	€ -	€ 6.500	€ 0	€ 6.500,00
EUR	Open access publications				€ 1.500	€ -	€ -	€ -	€ 500,00	€ -	€ 500,00	€ -	€ 500,00	€ -	€ 500,00	€ -	€ -	€ 1.500	€ 0	€ 1.500,00



**Gemeente
Rotterdam**

Maatschappelijke Ontwikkeling
Preventie en Welzijn

Bezoekadres: Het Timmerhuis
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Internet: www.rotterdam.nl

Van: S. Kuiper
Telefoon: 14 010
E-mail: s.kuiper@rotterdam.nl
c.c. w.jansen@Rotterdam.nl

Onderwerp: Letter of intent

Datum: 14 maart 2022

Retouradres: Postbus 70012, 3000 KP ROTTERDAM

Per email

Samantha Copeland,
TU Delft
Faculty of Technology, Policy and
Management

Dear dr. Samantha Copeland,

I, Sabine Kuiper, in my capacity of manager of the Department Prevention and Wellbeing at the city of Rotterdam hereby confirm that the Municipality of Rotterdam intends to contribute to the Healthy Neighbourhoods project.

The city of Rotterdam intends to collaborate within the network of experts from Erasmus University, Erasmus Medical Center and TU Delft and contribute to achieving the aims of this proposal:

Our team and project output will contribute directly to the main goals of Convergence: life-long health and well-being by supporting both health-conducive relationships with community members and institutions, as well as integrating care and support into the living, built, urban environment; attending to socio-economic equality in health and wellbeing by reducing disparities at the level of the neighbourhood through design by tailoring technologies introduced into communities to local needs; implementing and refining technologies that diagnose and recommend preventative measures for health and wellbeing throughout the lifecycle. These goals will be accomplished by tailoring technology to individuals and local stakeholders, as we develop principles for identifying and addressing the unique needs of different neighbourhoods, to implement technologies in ways that generate healthy neighbourhoods as well as healthy and self-sufficient individuals.



The municipality of Rotterdam intends to provide a € 125K in kind contribution to the project representing supervision and research expertise.

Yours sincerely,

Sabine Kuiper
Manager Department Prevention and Wellbeing

Date 11 March 2022
Our reference 22-113/SR/nvdh
Telephone +31 15 27 88779
E-mail s.roeser@tudelft.nl
Subject Letter of support



Faculty Technology, Policy and Management
Department Values Technology and
Innovation

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2628 BX Delft
Postal address
P.O. Box 5015
2600 GA Delft
The Netherlands

LETTER OF SUPPORT
for the
Tailoring Technologies for a Healthy Neighbourhood
PROJECT

I, Prof. Dr. Sabine Roeser, in my capacity of Head of the Department of Values, Technology, and Innovation at Delft University of Technology hereby confirm that we are aware of and support the flagship initiative *Tailoring Technologies for a Healthy Neighbourhood*, as applied for by the main applicant, Dr. Samantha Copeland, assistant professor at Delft University of Technology, TPM faculty.

Yours sincerely,

A handwritten signature in black ink, appearing to read "Sabine Roeser", positioned above a horizontal line.

Name: Prof. Dr. Sabine Roeser

Position: Head of the Department of Values, Technology, and Innovation at TPM faculty,
Delft University of Technology

Date: March 11, 2022